RAUNAK MONDAL

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EDUCATION

Carnegie Mellon University, School of Computer Science Aug 2025 – May 2028 (expected)
B.S. Computer Science, Machine Learning concentration · Pittsburgh, PA
Honors: 1st (Prelim), CMU March Madness ML · 5th, CMU AI Safety Estimathon · National Merit Finalist · HMMT 21st (Guts) · ARML Top 50 · Stanford SUMaC
Coursework: Computer Systems (15-213), Parallel & Sequential Data Structures and Algorithms (15-210), Data Structures & Algorithms (15-122), Functional Programming (15-150), Computer Security (15-330), Probability Theory for Computer Scientists (36-218), Concepts of AI (07-180)
Teaching: TA, Great Practical Ideas for Computer Scientists (07-131), the Unix and developer-toolchain course taken by ∼80% of first-year SCS students · Facilitator, AI Safety × Philosophy reading group, Carnegie AI Safety Initiative

EXPERIENCE

Twinly · Software Engineering Intern May 2026 – Present
AI desktop agent startup (macOS and Windows) · C#/.NET 9, Electron + React, Avalonia · 30+ PRs, 289 commits

- Shipped the plugin SDK: plugins declare tools by manifest with a JSON Schema and scoped permissions, the registry enforces at load that each tool's permissions are a subset of the manifest's, and an absent consent resolver defaults to an empty grant, so forgetting to wire consent blocks side-effect tools rather than opening them.
- Led a defensive security audit across 7 attack surfaces, authored a 24-finding severity-ranked report, and landed fixes across 12 files in C#, TypeScript, Python and PowerShell, including bearer-token auth on an unauthenticated sidecar. Audited the macOS build with gitleaks and trufflehog and hardened its CcompanionServer remote-desktop surface behind a RemoteControlSession gating design.
- Built PromptLab, the prompt A/B harness: FNV-1a 64-bit bucketing (`string.GetHashCode()` is randomized per process in .NET Core and would reshuffle every user on restart) with a two-sided pooled two-proportion z-test for rollup.

ScottyLabs × CMU Graduate Student Assembly · Backend & Database Developer Oct 2025 – Present
Student housing platform · PostgreSQL/Supabase, TypeScript, React · 240 PRs, 512 commits

- Own the database layer of a production rental-review platform: 85 RPC functions over 57 migrations and ∼19,000 lines of SQL, on a two-schema design where raw submissions land in a private admin schema nothing public reads and every public path returns through a SECURITY DEFINER RPC that aggregates and authorizes first.
- Found and fixed two access-control defects in a self-directed audit (an IDOR in an RPC that trusted a caller-supplied user ID, a missing role check on a publish path) and two rent-projection bugs from user reports: backward extrapolation on leases not yet renewed, and a COALESCE null/zero conflation inflating single-review averages.

Smartbite AI · Founding Engineer Jan 2025 – Present

- Built the core product from scratch; contributed to the CES 2026 live demo: LLM agent workflows pairing a recommender with nutritional constraint solving.

RESEARCH

Image Memorability with a Calibrated Trust Layer Oct 2026
independent project · PyTorch, Modal GPU

- Frozen CLIP ViT-B/32 features + ridge predict LaMem memorability at Spearman 0.699 (10k test), reaching the human split-half consistency ceiling ($\rho \approx 0.68$), over a ResNet18 baseline 0.619 ± 0.009 and a 0.097 brightness floor (fine-tuned MemNet 0.64). Split-conformal calibration over a heteroscedastic variance head gives near-exact coverage (0.905 at nominal 0.90); selective prediction cuts confident-slice MAE $0.069 \rightarrow 0.058$, and OOD distance routes out-of-domain ad creative at AUROC 0.93 where the model's own variance cannot. Reframed as creative A/B selection, the model picks the more memorable of two images 74% of the time, and gating on the prediction margin holds 87% on the confident half it auto-decides while deferring the close calls.

Do LLM Agents Run Controlled Experiments? (CRN-Bench) Jan 2026 – Present
advised by Prof. Amarnath Gupta, UC San Diego

- Benchmark for whether an agent probing a stochastic simulator holds the seed fixed across single-parameter changes or reads noise as signal. Scripted baselines bound the range before any model is evaluated: drawing fresh seeds confabulates 0 of 15 findings at a 50-call budget and 6 of 15 at 200, because argmax-then-t-test selection reaches significance without becoming real. Motivating case is a 21-parameter simulator I built over the OpenAlex citation graph, where screening found 15 parameters inert, one of them surfaced to users as a live control.

Reported vs. True Solution Quality in SkyDiscover Aug 2026
UC Berkeley's LLM-driven evolutionary search framework · independent research

- Dead-parameter injection that perturbs the containerized evaluator's score in train mode only, with the framework's own test mode as the held-out check. Depth 15, three runs per condition: control gap 0.000 in all three runs, injected condition +0.089, and injected runs' true quality peaked at iteration 8 while reported quality kept climbing. Also found two upstream defects: SEARCH/REPLACE diffs writing conflict markers into candidates, and an unseeded `random.sample`.

Do Less Coercive Interventions Reduce AI Deception? Fall 2026
SPAR Fall 2026 (Kairos), mentored by Aashiq Muhamed, ML PhD student, CMU

- Selected from a pool of over 6,000 applications. Model organisms with a planted goal conflict test whether safety interventions reduce deception or push it into hidden channels; deliverable is a co-authored paper plus released code or a small benchmark.

Stochastic Bandits: Regret and Allocational Stability Jun – Jul 2026
independent research

- Tested arXiv:2602.07472's regret-stability tradeoff numerically; at 8,000 replications per point the product is flat in T (fitted exponent -0.02, 95% CI $[-0.030, -0.010]$) where the paper predicts a log. Reported to the authors, who confirmed the reading.

One-Sided Jensen's Trace Inequality Jul 2026 – Present
with Zhiyu Zhang (Zhejiang University), an author of arXiv:2509.08911

- Proved the paper's open even-monomial conjecture $\Phi(x) = x^{2k}$ for $k = 3$ to 7 in all dimensions (previously only 2×2, $k = 3$) via Hermitian-unitary reduction, closed-form coefficients from closed-walk counting, and Sturm/resultant positivity certificates. Invited to co-lead the write-up.

OPEN SOURCE

- **nautechsystems/nautilus_trader** (28k stars; production Rust trading engine) · **#4934, merged:** SortinoRatio built a downside deviation from one daily observation and returned a finite, plausible number where SharpeRatio returns NaN through the $n < 2$ rule; added the same guard after daily downsampling so the two risk statistics agree on when dispersion is undefined, with rstest cases; the negative one fails on develop. **#4936, open:** on restart, Interactive Brokers replays execDetails through a process-local instrument provider that starts empty, dropping fills on instruments the cache already held (ten fills, three phantom positions over three restarts); widened the pre-reconciliation preload to every instrument a cached order references, in one batched call.
- **ccxt/ccxt** (44k stars; unified trading API over 100+ exchanges) · **#30288, landed on master:** a Python transpiler regex rewrote a leading `!` inside string literals into `not`, so two shipped WebSocket subscriptions were silently wrong. Added a lookbehind so the rule cannot fire after a quote and regenerated all Python output; the diff against the committed tree is exactly the three affected lines.
- **dlt-hub/dlt** (Python data-load tool) · **#4542, open:** the stored schema hash was persisted after a partial migration (`only_tables`), letting an interrupted load replay as complete and skip the remaining tables; submitted a fix gating the hash write on a full migration with regression tests across the DuckDB and SQLAlchemy destinations.
- **web-infra-dev/rspack** (13k stars; ByteDance's Rust bundler) · **#15300:** the minimize persistent cache keyed entries on an 8-byte hash and stored no identity on the value, so recovery could emit one asset with another's minimized body and no error. Strengthening the key cannot close it, since the association is what is wrong, so I added EntryIdentity to the persisted entry and treat a mismatch as a miss.
- **bytedance/bolt** (C++ Spark execution engine) · four open PRs (shiftrightunsigned, bround with HALF_EVEN via BigDecimal::setScale, octet_length, a macOS Conan build fix from an issue I filed), cross-checked against Apache Spark 3.5.5. **paradedb/paradedb** (9k stars; Postgres over DataFusion) · **#5978:** TableProvider::constraints() on the BM25 scan, which surfaced a latent group-by expansion bug. **OpenAPITools/openapi-generator** (27k stars) · three merged Kotlin generator fixes in 7.25.0 (#24742, #24756, #24725).

PUBLICATIONS & PRIOR RESEARCH

Mondal, R. et al. "Which Sequence-Based Neural Network Architecture Yields Optimal Performance in Classification of Autism-Related Behavior." AMIA 2024 (first author). · Mondal, R. et al. "Are Robots Ready to Deliver Autism Inclusion?" ACM CHI 2024 (co-author). · University of Hawai'i Digital Health Lab, Student Researcher (2023) · UC San Diego CSE, Research Intern (2022–2023).

TECHNICAL SKILLS

Languages: Python, C, C#, Rust, Java, TypeScript/JavaScript, SQL, Standard ML **Frameworks:** .NET 9, React, Electron, Node.js, Flask, D3.js
ML & Statistics: PyTorch, scikit-learn, NumPy, SymPy · probability, hypothesis testing, Monte Carlo simulation, bootstrap confidence intervals
Data & Infrastructure: PostgreSQL (PL/pgSQL, PostGIS, pg_trgm, pg_cron, RLS), Supabase, Neo4j, Docker, Git, Linux, Vitest, Playwright, xUnit